

Self-Reflection Report: UROP 3200 Research Experience

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1 Overview

Through this UROP project on LLM agent architectures for multi-agent coordination, I gained invaluable hands-on research experience that transcended traditional classroom learning. What began as an inquiry into whether large language models could play the Chinese card game Dou Di Zhu evolved into a comprehensive study of agent reasoning strategies, experimental methodology, and the challenges inherent in asymmetric multi-agent environments.

2 Learning and Growth

2.1 Technical Skills Development

Prior to this project, my experience with large language models was primarily limited to using them as productivity tools. This research forced me to understand LLMs as systems to be evaluated and compared—not merely utilized. I learned to design prompting strategies that elicit desired behaviors, implement tool-augmented architectures that extend LLM capabilities, and construct experimental frameworks that produce statistically meaningful results.

The implementation of four distinct agent architectures (Baseline, Chain-of-Thought, In-Context Learning, and Tool-Augmented) taught me how different reasoning approaches yield dramatically different outcomes. Watching Tool-Augmented agents achieve 90% win rate while In-Context Learning agents struggled at 56.7% demonstrated that architectural decisions matter profoundly in AI system design.

2.2 Research Methodology

I gained practical experience with the full research cycle: literature review, hypothesis formation, experimental design, data collection, analysis, and report writing. The 120+ games of data collected across experimental conditions provided sufficient samples to identify meaningful performance differences between agent types.

Perhaps most importantly, I learned to expect unexpected results. The poor performance of In-Context Learning agents—no improvement over Baseline, and significantly worse error rates—contradicted my initial hypothesis that static strategy guides would at least marginally improve performance. This taught me that research reveals truth regardless of our expectations.

3 Challenges Encountered

3.1 Asymmetric Role Coordination

The most intellectually challenging aspect of this project was designing experiments for an asymmetric game where one player competes against two collaborators. Traditional machine learning benchmarks often assume symmetric roles or two-player zero-sum environments. Developing evaluation methodology that fairly assessed agent performance across both landlord and farmer roles required careful consideration of rotating role designs and stratified win rate metrics.

3.2 Interpreting Error Rates

When In-Context Learning agents exhibited 193.47% error rate (compared to Baseline's 72.80%), I initially suspected implementation bugs. However, after extensive debugging, I concluded that static strategy guide retrieval introduced cognitive overhead that disrupted gameplay timing. The strategy guides—designed to help agents—actually hindered them by requiring time-consuming retrieval from extensive documentation. This reinforced my understanding that more information is not always better, and that system latency can fundamentally alter performance.

3.3 Academic Writing

Translating technical findings into clear academic prose proved more challenging than anticipated. The course requirement for 8-10 pages demanded precision in explanation while maintaining depth of analysis. Learning to present complex experimental results through effective tables and figures was an essential skill developed through multiple revision cycles.

4 Discoveries and Insights

4.1 LLM Agents as Game Players

The 100% landlord win rate achieved by Tool-Augmented agents surprised me. I had entered this project skeptical that LLMs could master asymmetric competitive roles, expecting at best parity with human players. The systematic dominance of tool-augmented approaches demonstrated that external tool validation effectively compensates for LLM limitations in strict rule-following and move validity checking.

4.2 Transparency versus Performance

The finding that Chain-of-Thought agents (80% win rate) lag behind Tool-Augmented agents (90% win rate) but provide transparent, verifiable reasoning chains raises important questions about the

trade-offs in AI system design. For high-stakes applications where decisions must be explainable, Chain-of-Thought may be preferable despite lower performance. This insight extends beyond card games to domains like medical diagnosis or autonomous vehicles where interpretability matters as much as accuracy.

4.3 Limitations of Static Retrieval

The failure of In-Context Learning suggests fundamental limitations in static knowledge retrieval for dynamic decision-making tasks. Card game play requires real-time adaptation to changing game states; prepended strategy guides cannot anticipate all possible situations. This contrasts with tool-augmented approaches that dynamically query relevant information at decision time.

5 Conclusion

This UROP project transformed my understanding of research from an abstract concept into a concrete experience of discovery, frustration, revelation, and growth. I emerged with deeper technical knowledge of LLM agents, practical research skills, and appreciation for the complexities of multi-agent systems.

The project demonstrated that LLM agents *can* master asymmetric competitive roles when appropriately architected, while also revealing limitations in reasoning-based approaches that require further investigation. These findings inform future work on hybrid architectures and extension to other multi-agent environments.

Beyond the technical outcomes, this research experience solidified my interest in artificial intelligence and prepared me for graduate-level research. The combination of theoretical understanding and practical implementation skills will serve as a foundation for future work in intelligent systems and multi-agent coordination.